Semantic and temporal review - consolidated V2

DeepSeek 4810 + CodeLlama 4309 | original figures retained; new discussion and 4309 analyses appended

Purpose

An internal evidence review for choosing what belongs in the paper and slides. This is not a finished manuscript narrative. Existing results, counterexamples and limitations are intentionally retained.

What is new

Comparison of all 27 changed CodeLlama target outputs with six shuffled-direction arms; structural hypotheses on natural base/merged texts; temporal localization across 50 regressions; return-newline preference; eight activation examples; all seven return-only cases, including unchanged and worsened implementations.

How to read older pages

The first 14 original pages are preserved without modification. Where later pages add nuance, use the newer analysis: return-newline preference is not error specificity, and continuous steering does not establish mediation by late natural activation.

Data scope

DeepSeek: 119 contamination improvements; CodeLlama: 50 logic/runtime regressions. Main report doses are 0 to 5. No new inference, generation, training or benchmark evaluation was performed for this compilation.

Reading guide and reproducibility

All page ranges refer to this consolidated PDF.

Section	Pages
Original visual review (preserved)	3-16
Updated findings and complete change tables	17-21
Exploratory behavior hypothesis figures	22-24
Uniform target / shuffle comparison	25-25
Temporal regions, return-newline statistics and eight examples	26-28
Code before / after and unchanged counterexamples	29-31

All 50 annotated programs

reports/CODELLAMA_4309_ALL_50_ANNOTATED_PTBR.md contains complete code with => / <= error regions, * first-activation marker, original line numbers, rationale and evaluator evidence.

Natural capture versus intervention baseline

Frozen captured code differs from currently re-extracted alpha-zero code in some cases. Activation anchors are aligned by exact matching blocks. Offline shared-latent activation is not an online per-model measurement; prompt activations were excluded.

Controls and experimental status

Three negative shuffled arms form the primary same-sign comparison; positive arms are sensitivity checks. Random CrossCoder latents are distinct from shuffled decoder coordinates. Current interpretations are exploratory on selected cohorts, without independent held-out validation.